# Graphing Quadratic Functions: Domain & Range

Notes · Summary · Practice Problems · Full Answer Key

**Name:** _____ **Date:** _____

*Estimated time: 40–55 minutes*

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## 1. Summary

A **quadratic function** has the form $y = ax^2 + k$ (with $a \neq 0$). Its graph is a **parabola** — a U-shaped curve.

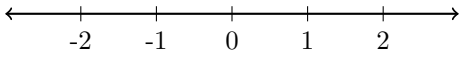
**Direction.** If $a > 0$ the parabola opens *up*; if $a < 0$ it opens *down*. The larger $|a|$ is, the *narrower* the parabola; the smaller $|a|$ is, the *wider* it is.

**Vertex.** For $y = ax^2 + k$ the vertex (turning point) is at $(0,\ k)$. This is the lowest point when the parabola opens up, and the highest point when it opens down.

**Domain** = the set of all allowed $x$-values (inputs).
**Range** = the set of all resulting $y$-values (outputs).

**Key facts for $y = ax^2 + k$:**
- **Domain: all real numbers**, $(-\infty, \infty)$ — you may square any number.
- If $a > 0$: **Range** is $y \geq k$, i.e. $[k, \infty)$.
- If $a < 0$: **Range** is $y \leq k$, i.e. $(-\infty, k]$.

**How do we find the domain?** Use a number line and ask: can $x$ be zero? negative? positive? For a plain parabola the answer is *yes* to all three, so the domain is all real numbers.




## 2. Worked Examples

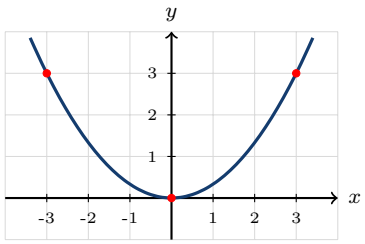
**Example — graph $y = \frac{1}{3}x^2$**

Build a table of values. Remember $x^2$ means $x \cdot x$, so a negative input gives a positive output:

$$y = \tfrac{1}{3}(-3)^2 = \tfrac{1}{3}\,(-3)(-3) = \tfrac{1}{3}(9) = 3.$$

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| $x$ | $y$ |
|---|---|
| $-3$ | 3 |
| 0 | 0 |
| 3 | 3 |




Since $a = \frac{1}{3} > 0$ the parabola opens up with vertex $(0, 0)$.

Domain: all real numbers $(-\infty, \infty)$ Range: $y \geq 0$, $[0, \infty)$

**Example — a downward, shifted parabola $y = -x^2 + 2$**

Here $a = -1 < 0$, so it opens down; the vertex is $(0, 2)$, the *highest* point.

Domain: all real numbers Range: $y \leq 2$, $(-\infty, 2]$

**Example — comparing the width of parabolas**

Compare $f(x) = -x^2$, $g(x) = 3x^2$, $h(x) = -\frac{1}{3}x^2$.

Line up the coefficients and take their **absolute values** — the sign only tells you which way the parabola opens, not how wide it is:

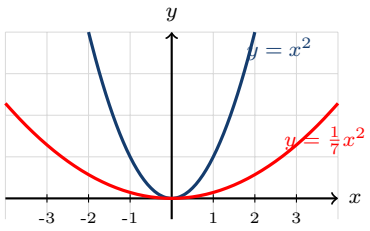
$$\text{coefficients: } -1,\ 3,\ -\tfrac{1}{3} \qquad \Longrightarrow \qquad |a|: \ 1,\ 3,\ \tfrac{1}{3}.$$

**Width rule.** When comparing the width of parabolas, use only $|a|$ (the absolute value of the coefficient):

- The **greater** $|a|$ is, the **narrower** the parabola.
- The **smaller** $|a|$ is, the **wider** the parabola.

So from narrowest to widest: $g$ ($|a| = 3$), then $f$ ($|a| = 1$), then $h$ $\left(|a| = \frac{1}{3}\right)$.

*Still confused?* Graph $y = \frac{1}{7}x^2$ next to $y = x^2$ on the same axes. The small coefficient $\frac{1}{7}$ makes a much wider, flatter parabola — confirming: smaller $|a| \Rightarrow$ wider.




**Conclusion.** The width of a parabola $y = ax^2$ depends only on $|a|$, not on its sign. A **greater** $|a|$ gives a **narrower** graph; a **smaller** $|a|$ gives a **wider** graph. The sign of $a$ only decides whether the parabola opens up or down.

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**Example — vertical shifts $y = ax^2 + c$**

**How is the graph of $y = 3x^2 + 2$ different from the graph of $y = 3x^2$?**

Make a table and compare the two columns — every $y$-value is exactly 2 larger:

| $x$ | $y = 3x^2$ | $y = 3x^2 + 2$ |
|---|---|---|
| $-2$ | 12 | 14 |
| $-1$ | 3 | 5 |
| 0 | 0 | 2 |
| 1 | 3 | 5 |
| 2 | 12 | 14 |

The parabola $y = 3x^2 + 2$ has the *same shape* as $y = 3x^2$, but every point is lifted straight up 2 units. **It is shifted 2 units up.**

**Vertical-shift rule.** Adding $c$ to $y = ax^2$ slides the whole parabola up or down without changing its shape:

- If $c > 0$: shift **up** $c$ units; the vertex moves to $(0, c)$.
- If $c < 0$: shift **down** $|c|$ units; the vertex moves to $(0, c)$.

**Example — solving $x^2 = k$ by square roots**

To solve $x^2 = 16$, take the square root of *both sides*. The key is that $\sqrt{x^2} = |x|$ (not just $x$), because squaring erases a negative sign:

$$x^2 = 16 \implies \sqrt{x^2} = \sqrt{16} \implies |x| = 4 \implies x = \pm 4 \quad \text{(i.e. } x = 4 \text{ or } x = -4\text{)}.$$

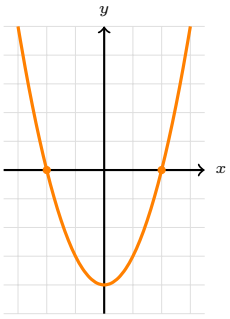
That is why a square-root equation has **two** solutions: both 4 and $-4$ square to 16.

**Square-root rule.** If $x^2 = k$ with $k > 0$, then $x = \pm\sqrt{k}$ (two solutions). If $k = 0$, then $x = 0$ (one solution). If $k < 0$, there is **no real solution**, since a square is never negative.

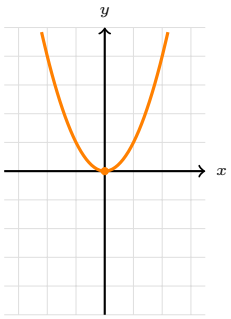
**Example — solving by graphing**

To solve an equation like $x^2 + c = 0$, graph the **related function** $y = x^2 + c$ and read off its **$x$-intercepts** (where $y = 0$). Each $x$-intercept is a solution.

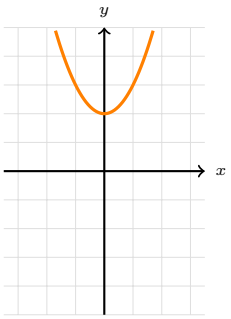
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**A.** $x^2 - 4 = 0$
$y = x^2 - 4$
Two solutions: $x = \pm 2$




**B.** $x^2 = 0$
$y = x^2$
One solution: $x = 0$




**C.** $x^2 + 2 = 0$
$y = x^2 + 2$
No real solution

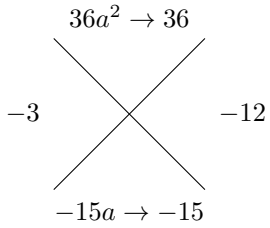
**Solving by graphing.** The real solutions of $ax^2 + c = 0$ are the $x$-intercepts of the related graph $y = ax^2 + c$. A parabola can cross the $x$-axis **twice** (two solutions), **touch** it once (one solution), or **miss** it entirely (no real solution).

**Example — factoring $ax^2 + bx + c$ (product–sum method)**

Factor $2a^2 - 15a + 18$. When the leading coefficient is not 1, use the **"X"** method:

1. Multiply the outer terms: $2a^2 \cdot 18 = 36a^2$ (top of the X). For simplicity we drop the $a^2$ and just use **36**.
2. The middle term is $-15a$; for simplicity we use $-15$ as the sum (bottom of the X). Find two numbers whose **product** is 36 and whose **sum** is $-15$. They are $-3$ and $-12$.
3. Split the middle term using those numbers and factor by grouping:

$$2a^2 - 15a + 18 = 2a^2 - 3a - 12a + 18 = a(2a - 3) - 6(2a - 3) = (2a - 3)(a - 6).$$




**Check:** $(2a - 3)(a - 6) = 2a^2 - 12a - 3a + 18 = 2a^2 - 15a + 18$. ✓

**Product–sum rule.** To factor $ax^2 + bx + c$: find two numbers that multiply to $a \cdot c$ and add to $b$. Use them to split the middle term, then factor by grouping.

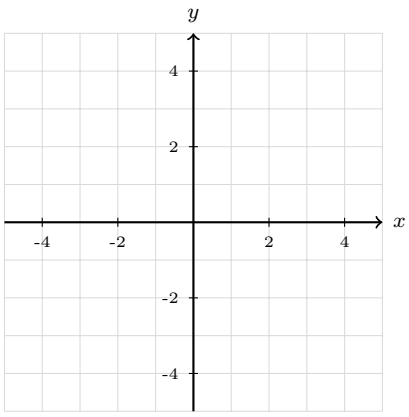
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## 3. Practice Problems

For each function: make a table of at least three points, sketch the parabola on the grid, and state the **domain** and **range**.

**Part A — Easy** *(basic $y = ax^2$)*

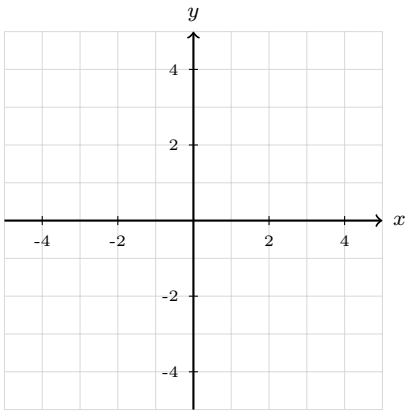
*Model.* $y = 3x^2$: points $(-1, 3), (0, 0), (1, 3)$; opens up, vertex $(0, 0)$. Domain: all reals $(-\infty, \infty)$; Range: $y \geq 0$, $[0, \infty)$.




**A1.** $y = x^2$

Domain: _____

Range: _____

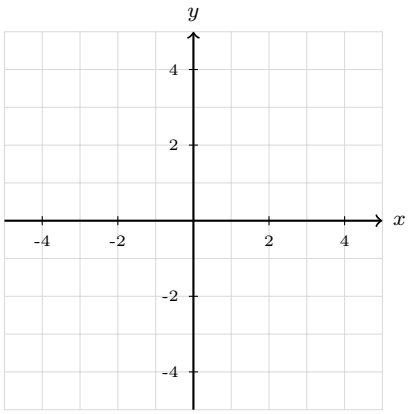



**A2.** $y = 2x^2$

Domain: _____

Range: _____

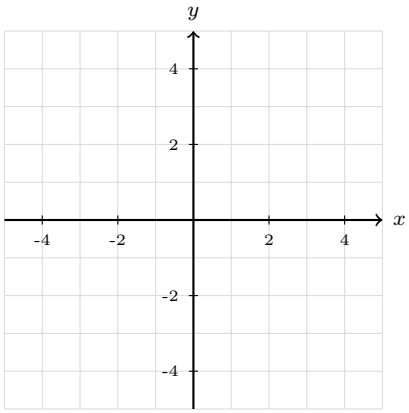
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**A3.** $y = \frac{1}{2}x^2$

Domain: _____

Range: _____




**A4.** $y = -x^2$

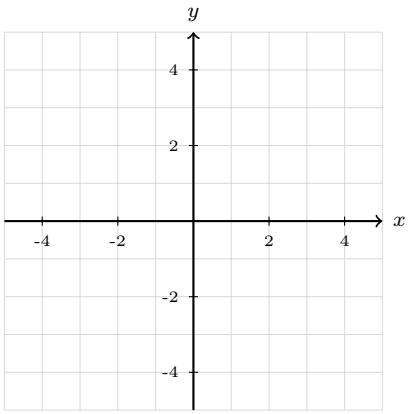
Domain: _____

Range: _____

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**Part B — Medium** *(vertical shifts and reflections $y = ax^2 + k$)*

*Model.* $y = 2x^2 - 3$: opens up (since $a > 0$), vertex $(0, -3)$. Domain: all reals; Range: $y \geq -3$, $[-3, \infty)$.

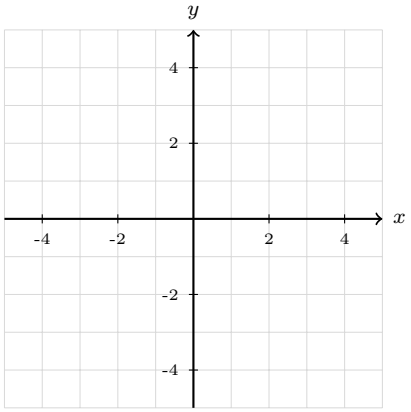



**B1.** $y = x^2 + 2$

Vertex: _____

Domain: _____

Range: _____



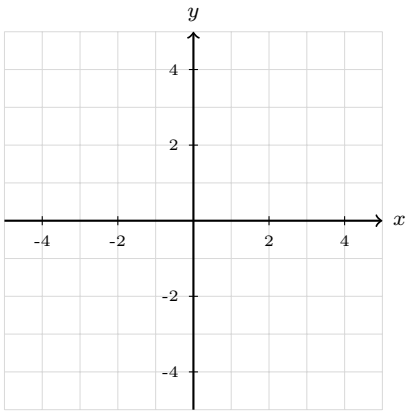

**B2.** $y = -2x^2$

Vertex: _____

Domain: _____

Range: _____

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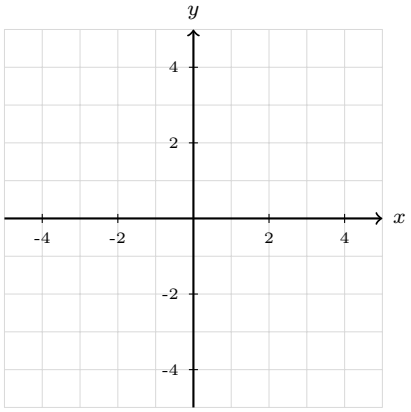



**B3.** $y = \frac{1}{3}x^2 - 1$

Vertex: _____

Domain: _____

Range: _____




**B4.** $y = -x^2 + 4$

Vertex: _____

Domain: _____

Range: _____

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**Part C — Hard** *(reasoning & applications)*

*Model.* State the domain and range of $y = 3x^2 - 2$ without graphing: $a = 3 > 0$ so it opens up; the constant $-2$ is the vertex $y$-value (a minimum). Domain: all reals; Range: $y \geq -2$, $[-2, \infty)$.

**C1.** Without graphing, state the domain and range of $y = -\dfrac{1}{2}x^2 + 3$. Explain how you knew the range from the equation alone.

**C2.** A parabola opens *upward* and its lowest point is $(0, -5)$. Write a possible equation, then give its domain and range.

**C3.** Two students disagree. One says the range of $y = x^2$ is "all real numbers" because the domain is all real numbers. Explain why that is wrong and give the correct range.

**C4. Application.** A ball is thrown and its height (in feet) after $t$ seconds is $h = -16t^2 + 4$. Physically, what values of $t$ make sense (the domain), and what is the greatest height the ball reaches (top of the range)?

**Part D — Comparing Width (uses $|a|$)**

*Model.* Order $y = 4x^2$, $y = x^2$, $y = \frac{1}{3}x^2$ from narrowest to widest: $|a| = 4$, $1$, $\frac{1}{3}$, so narrowest $\to$ widest is $4x^2$, then $x^2$, then $\frac{1}{3}x^2$.

**D1.** *(Easy)* Order these from **widest to narrowest** using only $|a|$:

$$f(x) = 5x^2, \qquad g(x) = x^2, \qquad h(x) = \tfrac{1}{4}x^2.$$

**D2.** *(Medium)* For $f(x) = -x^2$, $g(x) = 3x^2$, $h(x) = -\frac{1}{3}x^2$, write each $|a|$ and order the graphs

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from narrowest to widest. Does the negative sign change the width? Explain.

**D3.** *(Hard)* A parabola is *wider* than $y = x^2$ but *narrower* than $y = \frac{1}{10}x^2$, and it opens *down*. Write one possible equation and state its $|a|$.

**Part E — Vertical Shifts** $y = ax^2 + c$

For each, describe how the graph differs from the base parabola (shift direction and number of units), and give the new vertex.

*Model.* How is $y = 4x^2 - 1$ different from $y = 4x^2$? It is shifted **down 1** unit (same shape); new vertex $(0, -1)$.

**E1.** How is $y = 3x^2 + 4$ different from $y = 3x^2$?

**E2.** How is $y = x^2 - 5$ different from $y = x^2$?

**E3.** How is $y = -2x^2 + 1$ different from $y = -2x^2$?

**E4.** How is $y = \frac{1}{2}x^2 - 3$ different from $y = \frac{1}{2}x^2$?

**E5.** Graph $y = x^2$ and $y = x^2 + 2$ on the same axes. How are the two graphs related?

**Part F — Solving** $x^2 = k$ **by Square Roots**

Solve each equation. Give all real solutions (remember $x = \pm\sqrt{k}$).

*Model.* $x^2 = 64 \Rightarrow |x| = 8 \Rightarrow x = \pm 8$.

*Easy.* **F1.** $x^2 = 9$ **F2.** $x^2 = 49$

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*Medium.* **F3.** $3x^2 = 48$ **F4.** $x^2 = 20$

*Hard.* **F5.** $(x-1)^2 = 9$ **F6.** $x^2 = -16$

### Part G — Solving by Graphing

For each equation, graph the related function and state the solutions (how many, and what they are).

*Model.* $x^2 - 1 = 0$: the related graph $y = x^2 - 1$ crosses the $x$-axis at $-1$ and 1, so there are **two solutions**, $x = \pm 1$.

**G1.** $x^2 - 9 = 0$

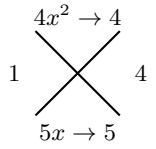
**G2.** $2x^2 + 8 = 0$

**G3.** $x^2 - 49 = -49$

### Part H — Factoring $ax^2 + bx + c$ (product–sum method)

Factor each trinomial completely. Check by multiplying back.

*Model.* Factor $2x^2 + 5x + 2$. The product of the outer terms is $2x^2 \cdot 2 = 4x^2$; for simplicity we drop the $x^2$ and use 4. The middle term is $5x$; likewise we use 5 for the sum. Find two numbers with product 4 and sum 5: 1 and 4.




$$2x^2 + 5x + 2 = 2x^2 + x + 4x + 2 = x(2x+1) + 2(2x+1) = (2x+1)(x+2).$$

*Easy.* **H1.** $2x^2 + 7x + 3$ **H2.** $3x^2 + 8x + 4$

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*Medium.* **H3.** $3x^2 - 10x + 8$ **H4.** $5x^2 + 13x + 6$

*Hard.* **H5.** $6x^2 - 7x - 3$ **H6.** $4x^2 - 4x - 15$

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## 4. Detailed Answer Key

### Part A — Easy

**A1.** $y = x^2$: opens up, vertex $(0, 0)$. Domain: all reals $(-\infty, \infty)$; Range: $y \geq 0$, $[0, \infty)$.

**A2.** $y = 2x^2$: narrower, opens up, vertex $(0, 0)$. Domain: all reals; Range: $y \geq 0$, $[0, \infty)$.

**A3.** $y = \frac{1}{2}x^2$: wider, opens up, vertex $(0, 0)$. Domain: all reals; Range: $y \geq 0$, $[0, \infty)$.

**A4.** $y = -x^2$: opens down, vertex $(0, 0)$. Domain: all reals; Range: $y \leq 0$, $(-\infty, 0]$.

### Part B — Medium

**B1.** $y = x^2 + 2$: opens up, vertex $(0, 2)$. Domain: all reals; Range: $y \geq 2$, $[2, \infty)$.

**B2.** $y = -2x^2$: opens down (narrow), vertex $(0, 0)$. Domain: all reals; Range: $y \leq 0$, $(-\infty, 0]$.

**B3.** $y = \frac{1}{3}x^2 - 1$: opens up (wide), vertex $(0, -1)$. Domain: all reals; Range: $y \geq -1$, $[-1, \infty)$.

**B4.** $y = -x^2 + 4$: opens down, vertex $(0, 4)$. Domain: all reals; Range: $y \leq 4$, $(-\infty, 4]$.

### Part C — Hard

**C1.** $a = -\frac{1}{2} < 0$ so the parabola opens down; the constant $+3$ is the vertex $y$-value, which is the maximum. **Domain:** all reals $(-\infty, \infty)$; **Range:** $y \leq 3$, $(-\infty, 3]$.

**C2.** Opens up with minimum $(0, -5)$, so $k = -5$ and any $a > 0$ works, e.g. $y = x^2 - 5$. **Domain:** all reals; **Range:** $y \geq -5$, $[-5, \infty)$.

**C3.** Domain and range are different questions. Squaring any real number gives a result $\geq 0$, so $y$ can never be negative. The correct **range** is $y \geq 0$, $[0, \infty)$ — not all reals.

**C4.** Time starts at release and ends when the ball lands ($h = 0$): $-16t^2 + 4 = 0 \Rightarrow t^2 = \frac{1}{4} \Rightarrow t = \frac{1}{2}$. Physical **domain:** $0 \leq t \leq \frac{1}{2}$ seconds. The parabola opens down with vertex at $t = 0$, so the greatest height is $h = 4$ feet.

### Part D — Comparing Width

**D1.** $|a|$ values are $5,\ 1,\ \frac{1}{4}$. Widest → narrowest (smallest $|a|$ first): $h(x) = \frac{1}{4}x^2$, then $g(x) = x^2$, then $f(x) = 5x^2$.

**D2.** $|a|$ values: $f: 1,\ g: 3,\ h: \frac{1}{3}$. Narrowest → widest: $g$ (3), $f$ (1), $h$ ($\frac{1}{3}$). The negative sign does *not* change width — it only flips the parabola to open downward. Width depends only on $|a|$.

**D3.** Need $\frac{1}{10} < |a| < 1$ and $a < 0$, e.g. $y = -\frac{1}{2}x^2$ with $|a| = \frac{1}{2}$. (Any $a$ between $-1$ and $-\frac{1}{10}$ works.)

### Part E — Vertical Shifts

**E1.** Shifted **up 4** units (same shape). New vertex $(0, 4)$.

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**E2.** Shifted **down 5** units. New vertex $(0, -5)$.

**E3.** Shifted **up 1** unit (still opens down). New vertex $(0, 1)$.

**E4.** Shifted **down 3** units. New vertex $(0, -3)$.

**E5.** Same shape; $y = x^2 + 2$ is $y = x^2$ shifted **up 2** units, vertex moving from $(0, 0)$ to $(0, 2)$.

**Part F — Solving $x^2 = k$ by Square Roots**

**F1.** $|x| = 3 \Rightarrow x = \pm 3$.

**F2.** $|x| = 7 \Rightarrow x = \pm 7$.

**F3.** $3x^2 = 48 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$.

**F4.** $x^2 = 20 \Rightarrow x = \pm\sqrt{20} = \pm 2\sqrt{5}$ (about $\pm 4.47$).

**F5.** $|x - 1| = 3 \Rightarrow x - 1 = \pm 3 \Rightarrow x = 4$ or $x = -2$.

**F6.** No real solution — a square can never equal a negative number.

**Part G — Solving by Graphing**

**G1.** $y = x^2 - 9$ crosses the $x$-axis at $-3$ and $3$. **Two solutions:** $x = \pm 3$.

**G2.** $2x^2 + 8 = 0 \Rightarrow y = 2x^2 + 8$ has vertex $(0, 8)$ and opens up, so it never reaches the $x$-axis. **No real solution.**

**G3.** $x^2 - 49 = -49 \Rightarrow x^2 = 0$; $y = x^2$ touches the $x$-axis only at the origin. **One solution:** $x = 0$.

**Part H — Factoring $ax^2 + bx + c$**

**H1.** $ac = 6$, sum $7 \Rightarrow 1, 6$: $2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$.

**H2.** $ac = 12$, sum $8 \Rightarrow 2, 6$: $3x^2 + 2x + 6x + 4 = x(3x + 2) + 2(3x + 2) = (3x + 2)(x + 2)$.

**H3.** $ac = 24$, sum $-10 \Rightarrow -4, -6$: $3x^2 - 4x - 6x + 8 = x(3x - 4) - 2(3x - 4) = (3x - 4)(x - 2)$.

**H4.** $ac = 30$, sum $13 \Rightarrow 3, 10$: $5x^2 + 3x + 10x + 6 = x(5x + 3) + 2(5x + 3) = (5x + 3)(x + 2)$.

**H5.** $ac = -18$, sum $-7 \Rightarrow -9, 2$: $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$.

**H6.** $ac = -60$, sum $-4 \Rightarrow -10, 6$: $4x^2 - 10x + 6x - 15 = 2x(2x - 5) + 3(2x - 5) = (2x - 5)(2x + 3)$.

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